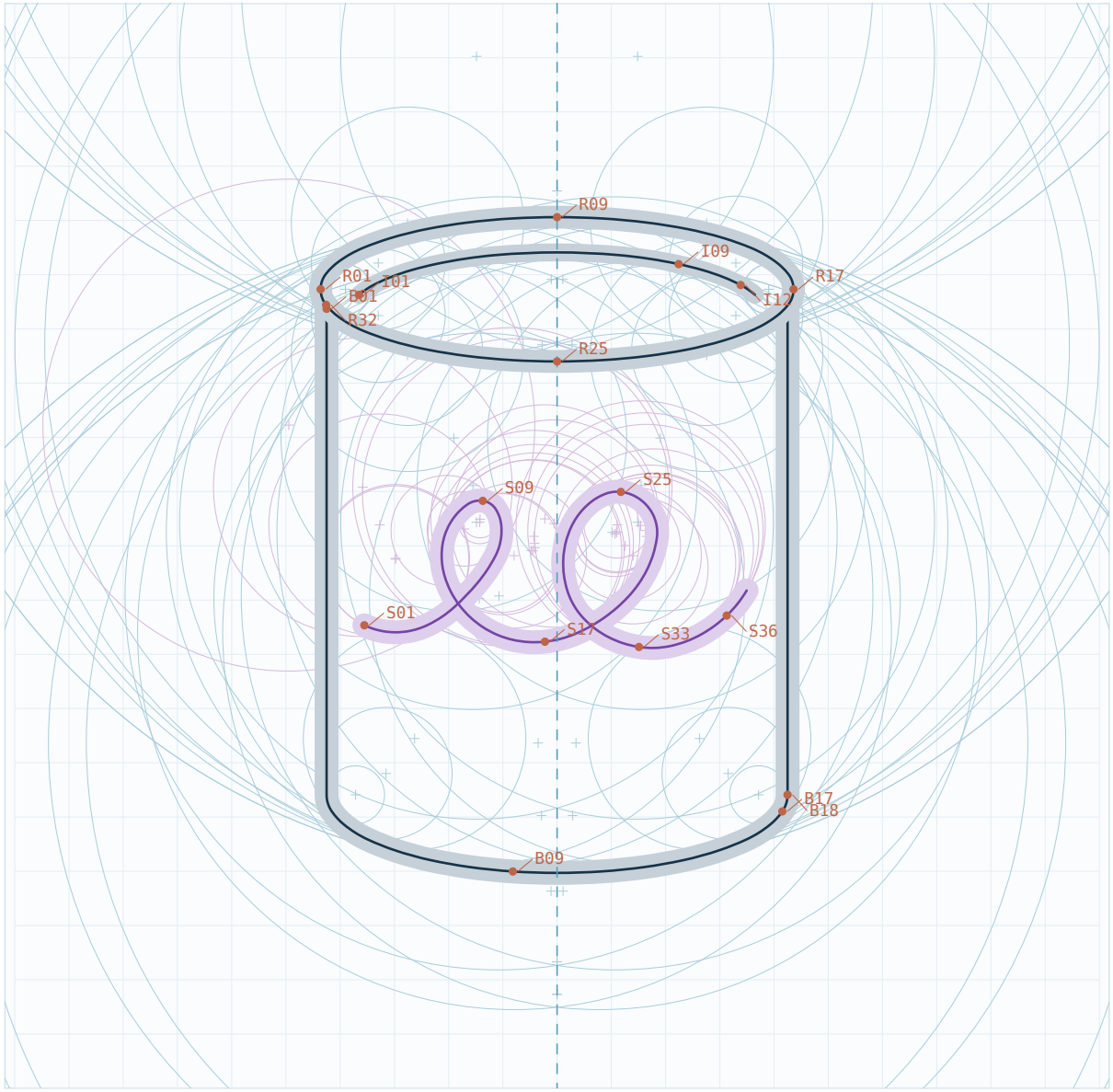
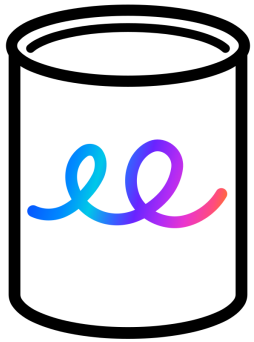
从图像，到可复建的几何原稿

原型保留 · 圆弧规范化 · 独立于原图的参数复建



这一稿是复建基准

把已确认图像转为直线、相切圆弧和确定线宽。辅助圆来自实际轮廓；每一段都能在参数表中找到圆心、半径、起终点及方向。

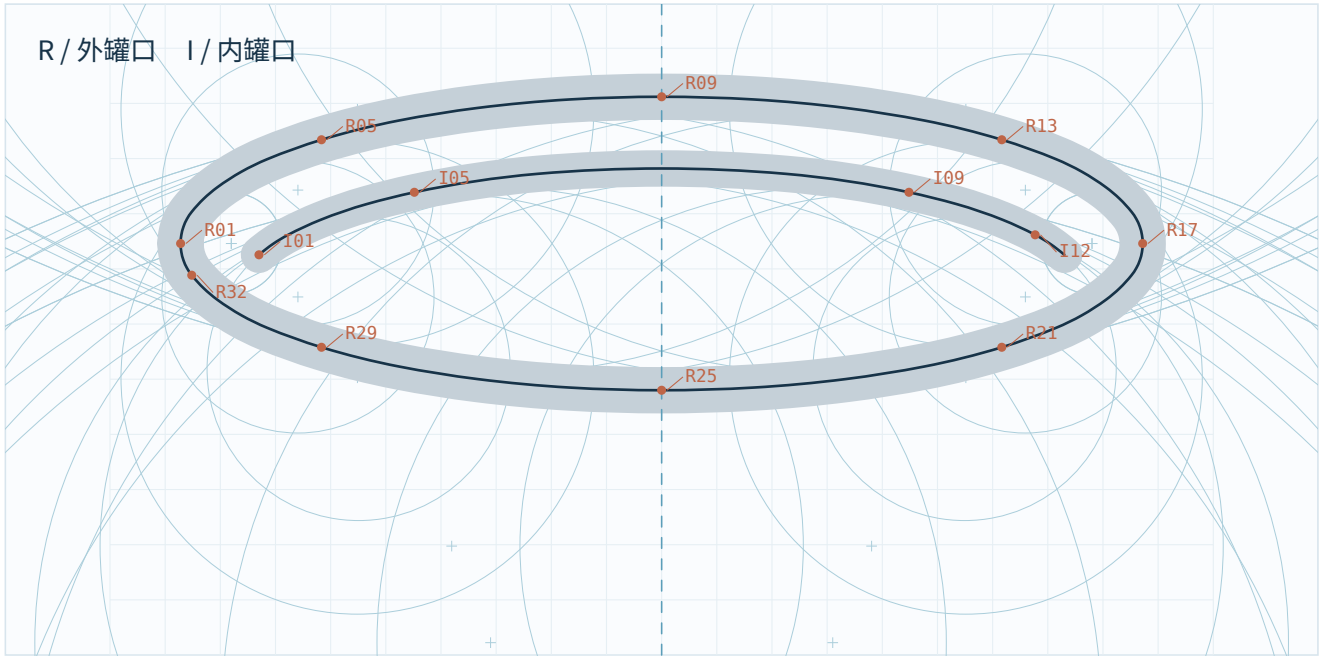


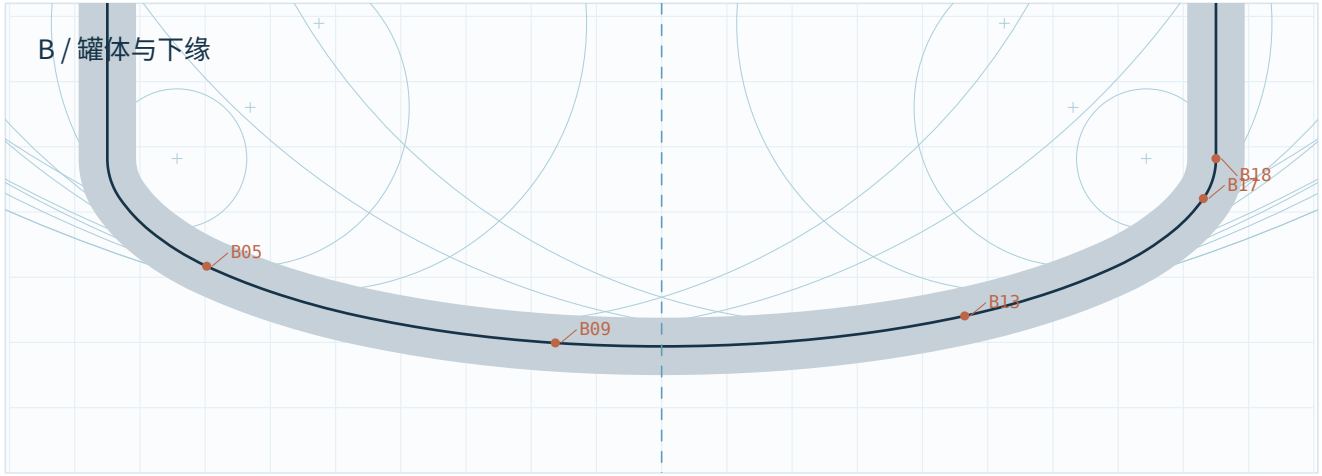
画板 / 坐标	1000 x 1000 U
几何元素	96 arcs + 2 lines
版本 / 复建基准	G1.0 / JSON + SVG

1:1 指本矢量规范版；生成位图的手工感细节经过微量规范化，不能称为逐像素无损转换。

罐体：中心线、圆弧与线宽

淡色带是实际描边范围，深色线是中心线；细圆为对应圆弧的完整母圆





基准尺寸

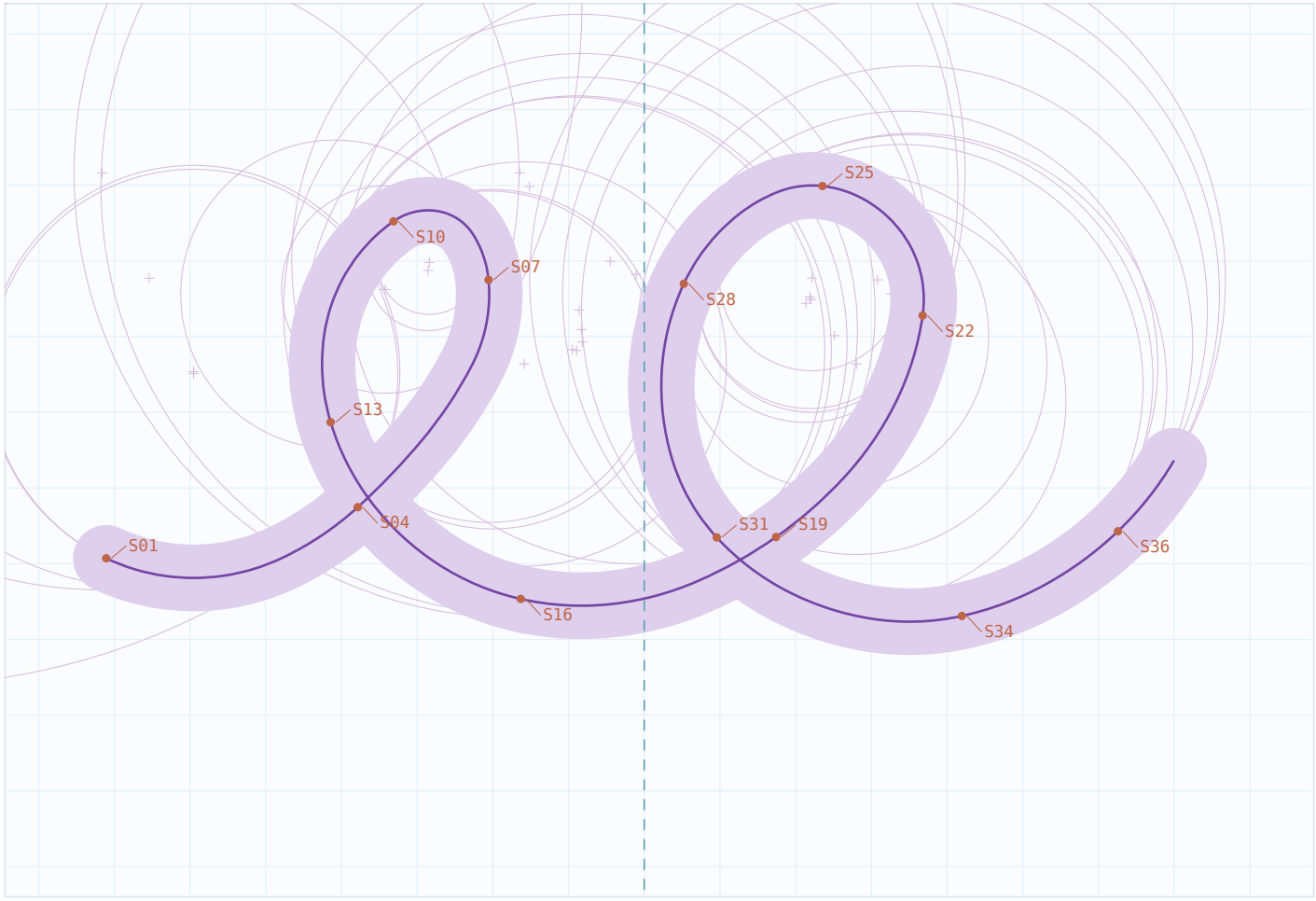
对称轴 x	500.000 U
左 / 右侧中心线	287.500 / 712.500
轮廓占用范围 x	271.500 ... 728.500
轮廓占用范围 y	186.500 ... 812.500
罐体 B 线宽	22.000 U
外罐口 R 线宽	21.000 U
内罐口 I 线宽	16.500 U

外罐口是闭合圆弧链。罐体从左上起，沿左侧向下，经底部回到右上。内罐口两端使用半径 8.25 U 的圆头。

绘制顺序：B → R → I → S。先按中心线绘制，再按指定宽度描边；无需徒手补边。

双圈：一条连续路径，36 段圆弧

沿 S01 至 S36 顺序连接；保留左右圈的不对称和手写节奏



左尾起笔 → 左圈 → 中间回落 → 右圈 → 右尾收笔

标注只显示部分节点以保持可读性；S01-S36 的完整参数在第 6 页与 CSV 中逐段列出。

复建规则

按表中的起点 P0、终点 P1、圆心 C 和半径 R 绘制各段圆弧，依次连接。圈内交叉处为同一笔画的并集，不增加立体遮挡。

笔画宽度	22.000 U
两端圆头半径	11.000 U
连接间隙	0.000000 U
端点圆误差	< 0.000001 U
切向偏差	< 0.00001 deg

所有接点均经过位置与切线连续性检查。数字复建直接读取 JSON，不用从图片中重新描摹。

圆弧参数 1 / 3 · 罐体与外罐口

B：线宽 22 U；R：线宽 21 U。直线段只使用 P0、P1，其余列留空。

ID	P0 起点 (x, y)	P1 终点 (x, y)	C 圆心 (x, y)	R 半径	θ0 起始角	Δθ 扫角	方向
B01	(287.500, 281.500)	(287.500, 729.500)	-	-	-	-	LINE
B02	(287.500, 729.500)	(292.329, 744.821)	(314.222, 729.500)	26.722	180.000	-34.985	CCW
B03	(292.329, 744.821)	(303.676, 757.053)	(342.269, 709.872)	60.955	145.015	-15.732	CCW
B04	(303.676, 757.053)	(325.580, 770.766)	(368.631, 677.645)	102.591	129.283	-14.471	CCW
B05	(325.580, 770.766)	(349.740, 780.412)	(428.265, 548.652)	244.701	114.811	-6.094	CCW
B06	(349.740, 780.412)	(383.827, 789.826)	(448.981, 487.512)	309.255	108.718	-6.555	CCW
B07	(383.827, 789.826)	(418.680, 796.019)	(486.242, 314.623)	486.115	102.162	-4.173	CCW
B08	(418.680, 796.019)	(459.241, 800.177)	(494.704, 254.323)	547.005	97.989	-4.272	CCW
B09	(459.241, 800.177)	(500.000, 801.500)	(500.000, 172.811)	628.689	93.717	-3.717	CCW
B10	(500.000, 801.500)	(540.759, 800.177)	(500.000, 172.811)	628.689	90.000	-3.717	CCW
B11	(540.759, 800.177)	(581.320, 796.019)	(505.296, 254.323)	547.005	86.283	-4.272	CCW
B12	(581.320, 796.019)	(616.173, 789.826)	(513.758, 314.623)	486.115	82.011	-4.173	CCW
B13	(616.173, 789.826)	(650.260, 780.412)	(551.019, 487.512)	309.255	77.838	-6.555	CCW
B14	(650.260, 780.412)	(674.420, 770.766)	(571.735, 548.652)	244.701	71.282	-6.094	CCW
B15	(674.420, 770.766)	(696.324, 757.053)	(631.369, 677.645)	102.591	65.189	-14.471	CCW
B16	(696.324, 757.053)	(707.671, 744.821)	(657.731, 709.872)	60.955	50.717	-15.732	CCW
B17	(707.671, 744.821)	(712.500, 729.500)	(685.778, 729.500)	26.722	34.985	-34.985	CCW
B18	(712.500, 729.500)	(712.500, 281.500)	-	-	-	-	LINE
R01	(282.000, 263.500)	(287.048, 249.133)	(304.967, 263.500)	22.967	180.000	38.721	CW
R02	(287.048, 249.133)	(298.594, 238.052)	(335.166, 287.712)	61.674	-141.279	14.910	CW
R03	(298.594, 238.052)	(321.232, 225.293)	(362.134, 324.330)	107.151	-126.370	13.929	CW
R04	(321.232, 225.293)	(345.851, 216.477)	(425.710, 478.270)	273.702	-112.440	5.476	CW
R05	(345.851, 216.477)	(380.891, 207.765)	(446.435, 546.212)	344.735	-106.964	6.004	CW
R06	(380.891, 207.765)	(416.575, 202.062)	(485.637, 748.640)	550.924	-100.960	3.759	CW
R07	(416.575, 202.062)	(458.205, 198.220)	(494.437, 818.283)	621.121	-97.201	3.857	CW
R08	(458.205, 198.220)	(500.000, 197.000)	(500.000, 913.484)	716.484	-93.344	3.344	CW
R09	(500.000, 197.000)	(541.795, 198.220)	(500.000, 913.484)	716.484	-90.000	3.344	CW
R10	(541.795, 198.220)	(583.425, 202.062)	(505.563, 818.283)	621.121	-86.656	3.857	CW
R11	(583.425, 202.062)	(619.109, 207.765)	(514.363, 748.640)	550.924	-82.799	3.759	CW
R12	(619.109, 207.765)	(654.149, 216.477)	(553.565, 546.212)	344.735	-79.040	6.004	CW
R13	(654.149, 216.477)	(678.768, 225.293)	(574.290, 478.270)	273.702	-73.036	5.476	CW

坐标单位 U；角度为度。原点左上，x 向右、y 向下，0° 向右；CW 顺时针，CCW 逆时针。

圆弧参数 2 / 3 · 外罐口与内罐口

R：线宽 21 U；I：线宽 16.5 U。首尾弧依序连接，闭合 R 环。

ID	P0 起点 (x, y)	P1 终点 (x, y)	C 圆心 (x, y)	R 半径	θ0 起始角	Δθ 扫角	方向
R14	(678.768, 225.293)	(701.406, 238.052)	(637.866, 324.330)	107.151	-67.560	13.929	CW
R15	(701.406, 238.052)	(712.952, 249.133)	(664.834, 287.712)	61.674	-53.630	14.910	CW
R16	(712.952, 249.133)	(718.000, 263.500)	(695.033, 263.500)	22.967	-38.721	38.721	CW
R17	(718.000, 263.500)	(712.952, 277.867)	(695.033, 263.500)	22.967	0.000	38.721	CW
R18	(712.952, 277.867)	(701.406, 288.948)	(664.834, 239.288)	61.674	38.721	14.910	CW
R19	(701.406, 288.948)	(678.768, 301.707)	(637.866, 202.670)	107.151	53.630	13.929	CW
R20	(678.768, 301.707)	(654.149, 310.523)	(574.290, 48.730)	273.702	67.560	5.476	CW
R21	(654.149, 310.523)	(619.109, 319.235)	(553.565, -19.212)	344.735	73.036	6.004	CW
R22	(619.109, 319.235)	(583.425, 324.938)	(514.363, -221.640)	550.924	79.040	3.759	CW
R23	(583.425, 324.938)	(541.795, 328.780)	(505.563, -291.283)	621.121	82.799	3.857	CW
R24	(541.795, 328.780)	(500.000, 330.000)	(500.000, -386.484)	716.484	86.656	3.344	CW
R25	(500.000, 330.000)	(458.205, 328.780)	(500.000, -386.484)	716.484	90.000	3.344	CW
R26	(458.205, 328.780)	(416.575, 324.938)	(494.437, -291.283)	621.121	93.344	3.857	CW
R27	(416.575, 324.938)	(380.891, 319.235)	(485.637, -221.640)	550.924	97.201	3.759	CW
R28	(380.891, 319.235)	(345.851, 310.523)	(446.435, -19.212)	344.735	100.960	6.004	CW
R29	(345.851, 310.523)	(321.232, 301.707)	(425.710, 48.730)	273.702	106.964	5.476	CW
R30	(321.232, 301.707)	(298.594, 288.948)	(362.134, 202.670)	107.151	112.440	13.929	CW
R31	(298.594, 288.948)	(287.048, 277.867)	(335.166, 239.288)	61.674	126.370	14.910	CW
R32	(287.048, 277.867)	(282.000, 263.500)	(304.967, 263.500)	22.967	141.279	38.721	CW
I01	(317.515, 268.617)	(330.734, 259.561)	(362.683, 320.370)	68.691	-131.113	13.396	CW
I02	(330.734, 259.561)	(345.358, 252.797)	(404.852, 400.629)	159.354	-117.718	5.796	CW
I03	(345.358, 252.797)	(366.363, 245.629)	(422.467, 444.400)	206.536	-111.922	6.160	CW
I04	(366.363, 245.629)	(387.924, 240.281)	(460.760, 580.067)	347.504	-105.762	3.663	CW
I05	(387.924, 240.281)	(443.649, 231.932)	(482.552, 681.728)	451.475	-102.099	7.155	CW
I06	(443.649, 231.932)	(500.000, 229.500)	(500.000, 883.463)	653.963	-94.943	4.943	CW
I07	(500.000, 229.500)	(556.351, 231.932)	(500.000, 883.463)	653.963	-90.000	4.943	CW
I08	(556.351, 231.932)	(612.076, 240.281)	(517.448, 681.728)	451.475	-85.057	7.155	CW
I09	(612.076, 240.281)	(633.637, 245.629)	(539.240, 580.067)	347.504	-77.901	3.663	CW
I10	(633.637, 245.629)	(654.642, 252.797)	(577.533, 444.400)	206.536	-74.238	6.160	CW
I11	(654.642, 252.797)	(669.266, 259.561)	(595.148, 400.629)	159.354	-68.078	5.796	CW
I12	(669.266, 259.561)	(682.485, 268.617)	(637.317, 320.370)	68.691	-62.282	13.396	CW

坐标单位 U；角度为度。原点左上，x 向右、y 向下，0° 向右；CW 顺时针，CCW 逆时针。

圆弧参数 3 / 3 · 双圈连续笔画

S：线宽 22 U；圆头、圆连接。表格保留 3 位小数；完整浮点参数见配套 JSON / CSV。

ID	P0 起点 (x, y)	P1 终点 (x, y)	C 圆心 (x, y)	R 半径	θ0 起始角	Δθ 扫角	方向
S01	(322.324, 573.218)	(351.359, 579.691)	(351.163, 512.221)	67.471	115.304	-25.471	CCW
S02	(351.359, 579.691)	(380.383, 573.106)	(351.161, 511.568)	68.123	89.833	-25.235	CCW
S03	(380.383, 573.106)	(405.384, 556.328)	(336.507, 480.710)	102.284	64.598	-16.927	CCW
S04	(405.384, 556.328)	(426.417, 534.419)	(252.703, 388.703)	226.737	47.671	-7.680	CCW
S05	(426.417, 534.419)	(443.031, 509.560)	(320.926, 445.930)	137.690	39.991	-12.466	CCW
S06	(443.031, 509.560)	(448.573, 481.251)	(397.859, 486.021)	50.937	27.524	-32.897	CCW
S07	(448.573, 481.251)	(443.622, 466.476)	(414.483, 484.458)	34.240	-5.373	-26.306	CCW
S08	(443.622, 466.476)	(431.502, 458.509)	(429.019, 475.487)	17.159	-31.679	-50.001	CCW
S09	(431.502, 458.509)	(417.183, 461.930)	(428.629, 478.157)	19.857	-81.680	-43.518	CCW
S10	(417.183, 461.930)	(405.793, 472.717)	(448.722, 506.642)	54.715	-125.198	-16.485	CCW
S11	(405.793, 472.717)	(394.331, 499.208)	(449.766, 507.467)	56.047	-141.683	-29.843	CCW
S12	(394.331, 499.208)	(396.434, 528.283)	(460.362, 509.046)	66.759	188.474	-25.221	CCW
S13	(396.434, 528.283)	(410.019, 554.870)	(476.257, 504.263)	83.359	163.253	-20.634	CCW
S14	(410.019, 554.870)	(431.967, 575.178)	(479.516, 501.772)	87.460	142.619	-19.685	CCW
S15	(431.967, 575.178)	(459.223, 586.640)	(477.687, 504.596)	84.096	122.934	-20.251	CCW
S16	(459.223, 586.640)	(488.811, 588.361)	(479.241, 497.694)	91.171	102.683	-18.708	CCW
S17	(488.811, 588.361)	(517.496, 580.803)	(478.559, 491.235)	97.666	83.974	-17.470	CCW
S18	(517.496, 580.803)	(543.530, 566.183)	(458.833, 445.859)	147.144	66.504	-11.646	CCW
S19	(543.530, 566.183)	(565.971, 546.542)	(462.087, 450.482)	141.490	54.858	-12.099	CCW
S20	(565.971, 546.542)	(582.849, 522.117)	(488.758, 475.145)	105.164	42.759	-16.230	CCW
S21	(582.849, 522.117)	(591.938, 493.043)	(497.331, 479.424)	95.583	26.529	-18.338	CCW
S22	(591.938, 493.043)	(584.881, 465.429)	(555.121, 487.743)	37.197	8.191	-45.053	CCW
S23	(584.881, 465.429)	(573.402, 455.151)	(553.426, 489.014)	39.315	-36.862	-22.601	CCW
S24	(573.402, 455.151)	(558.828, 450.274)	(554.673, 486.900)	36.861	-59.463	-24.063	CCW
S25	(558.828, 450.274)	(544.317, 452.149)	(555.380, 480.669)	30.590	-83.527	-27.673	CCW
S26	(544.317, 452.149)	(531.356, 459.513)	(562.761, 499.700)	51.002	-111.200	-16.807	CCW
S27	(531.356, 459.513)	(513.054, 482.550)	(570.082, 509.068)	62.892	-128.007	-27.055	CCW
S28	(513.054, 482.550)	(505.778, 511.375)	(585.183, 516.089)	79.545	-155.062	-21.540	CCW
S29	(505.778, 511.375)	(509.306, 540.818)	(589.103, 516.322)	83.472	-176.602	-20.463	CCW
S30	(509.306, 540.818)	(523.930, 566.307)	(572.815, 521.322)	66.434	162.935	-25.556	CCW
S31	(523.930, 566.307)	(547.388, 584.155)	(585.684, 509.480)	83.922	137.379	-20.229	CCW
S32	(547.388, 584.155)	(575.510, 593.233)	(589.289, 502.449)	91.823	117.150	-18.520	CCW
S33	(575.510, 593.233)	(604.898, 592.254)	(587.574, 513.750)	80.393	98.631	-21.075	CCW
S34	(604.898, 592.254)	(632.591, 581.802)	(582.629, 491.343)	103.340	77.556	-16.468	CCW
S35	(632.591, 581.802)	(656.431, 564.283)	(577.052, 481.245)	114.875	61.088	-14.797	CCW
S36	(656.431, 564.283)	(674.813, 541.141)	(581.483, 485.880)	108.463	46.291	-15.661	CCW

坐标单位 U；角度为度。原点左上，x 向右、y 向下，0° 向右；CW 顺时针，CCW 逆时针。

复建流程、渐变与验证

从参数重建，不依赖原位图；颜色由固定渐变轴与色标定义

按这五步重建

01 建立 1000 × 1000 U 画板；原点左上，x 向右、y 向下。输出边长为 L 时，1 U = L / 1000。

02 按表定位圆心 C，以 R 作圆；取 P0 至 P1 之间、符合 Δθ 方向的弧。直线段连接两端即可。

03 按 B → R → I → S 排列路径。线宽分别为 22、21、16.5、22 U，端点与连接均为圆形。

04 S 使用右侧渐变；罐体 #000000、背景 #FFFFFF。隐藏辅助线后导出。

05 也可直接运行源文件包中的 rebuild.py。只依赖标准 Python 与 JSON，可生成同一 SVG。

比例与印刷

SVG 已设置 100 mm 画板：1 U = 0.1 mm。PDF 不是 1:1 展开尺寸；按数字换算，勿拿页面缩放后的图形直接量长度。

实际长度校准： 10 mm（打印请关闭“适应页面”）

固定渐变定义

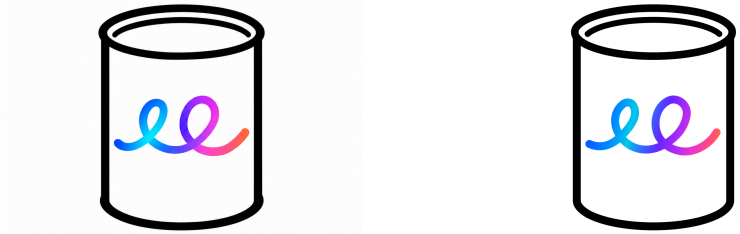
SVG userSpaceOnUse / sRGB / linear

P0 = (324, 510) P1 = (670, 605)



0%	18%	32%	45%	56%	70%	82%	94%	100%
#0754FF	#0089FF	#00C9E9	#3268FF	#6A24FF	#B32CF4	#EB35CF	#FF4C80	#FF6B3B

这是本规范版的渐变定义。原生成图的局部色彩经过参数化整理，不宣称与原图逐像素同色。



已确认位图　　圆弧规范版

在 1254 × 1254 像素下：轮廓平均偏差 0.878 px，95% ⩽ 2.236 px，最大 5.000 px。

测量方法：min(R,G,B)<180 的前景掩膜，计算双向边界距离；含抗锯齿、线宽归一与拟合差异。